



Sephadex Cleaning for Sanger Sequencing

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Background

This is a protocol to prepare a Millipore HV plate packed with Sephadex columns for post-sequencing reaction purification. Dry Sephadex powder is transferred to a 96-well Millipore plate, hydrated, and excess water is removed by centrifugation.

Helpful Hints or Important Considerations

- Sephadex absorbs moisture readily. Keep the Sephadex container tightly closed when not in use. Take Sephadex out of the white Sephadex container only when there is not enough Sephadex in the Nalgene bottle.
- Anything coated with dry Sephadex becomes slippery. Handle with care.
- After Sephadex plate preparation, clean up the Sephadex bench and make sure the toolbox is closed.
- Before each use, mark a Millipore HV plate to keep track of the times it has been used.
- All supplies for adding dry Sephadex to Millipore HV plate should be in Sephadex box near centrifuges (and are marked with an * below)

Materials & Reagents

- 100 ml clear Nalgene bottle with Sephadex*
- White Sephadex container*
- MultiScreen Column Loader*
- razor blade*
- weigh boat*
- Millipore HV plate
- Milli-Q ultrapure H₂O
- H₂O trough (use a clean plastic cover of tip refills)
- 200 µl Liquidator with P200 tips
- 20 µl Liquidator with P20 tips
- 96-well flat-bottomed microwell plate (found drying in rack between the sink and the Liquidator or on shelf above Sephadex station)
- Centrifuge (Beckman Coulter Allegra-X14)
- Clean PCR plate (found in box next to the fume hood outside the sequencing room)
- Grey Silicone Septa mat (found on the mini fridge-freezer at the Wolverine bench)

Protocol

Prepare Sephadex Plate

1. Pour Sephadex from the bottle onto a column loader close to the metal pin. Use the razor blade to move Sephadex and fill all wells. Scrape excess Sephadex with the razor into the weighing boat. When all plates have been filled, pour the excess Sephadex from the weighing boat back into the bottle.



2. Place a Millipore HV plate upside down over the column loader (i.e. have the open holes on the plate and loader facing each other), hold them together firmly, and quickly flip over. Pushing the Millipore plate against the pin helps keep both plates from sliding. With the column loader on top, tap plates on table while holding the plates firmly together. Sephadex is now transferred to the Millipore plate. Once all plates are loaded, return the Sephadex tools into the box and close the lid.
3. Transfer 300 μ l of deionized water to each filled Millipore HV plate using the 200 μ l Liquidator by pipetting 150 μ l twice.
4. Let Millipore plate sit for at least an hour to hydrate Sephadex.
5. Place the Millipore plate on top of a 96-well flat-bottomed microwell plate and centrifuge for 5 minutes at 2,100 G (Program 1 on the Beckman Coulter Allegra-X14). Discard the excess water in the microwell plate in the sink. The Sephadex plate is now ready to use.

Load Sephadex Plate

6. Label a clean PCR plate (found in box next to the fume hood outside the sequencing room) with your SI username and tracking ID number (the number given to the plate when you submit it for sequencing at <https://labreg.si.edu>). See printout above Wolverine fridge-freezer for an example of how to label this plate.
7. Place the prepared Sephadex plate on your labeled clean PCR plate. Make sure the well positions line up. Pipette 10 μ l of cycle sequencing reactions onto the prepared Sephadex plate using multichannel pipette or P20 liquidator, trying not to disturb Sephadex columns. Centrifuge for 5 minutes at 2,100 G (Program 1) to transfer the cycle sequencing product into the sequencing plate.
8. Dry the sequencing product on a thermocycler by heating at 94 °c for 15 minutes. Check to make sure the liquid has completely evaporated. Seal with a Silicone Septa mat and place it in the designated mini freezer at the Wolverine bench.

Clean Sephadex Plate

9. Remove used Sephadex columns by inverting the Millipore HV plate over a trash receptacle and tapping a few times. Rinse the top of the Millipore with deionized water.
10. Transfer 300 μ l of deionized water to the dirty Millipore HV plate with the Liquidator by pipetting 150 μ l twice.
11. Place the Millipore HV plate on top of a 96-well microwell plate and centrifuge for 5 minutes at 2,100 G (Program 1). Discard the water in the microwell plate.
12. Repeat steps 10 and 11
13. Dry the Millipore HV plate by leaving the lid open. It can be reused 10 times.

Additional Information/Appendices

- See Jeff or Carrie at the LAB Manager's office if there is no Sephadex container at the Sephadex bench.